SMART BATTERY MANAGEMENT SYSTEM

Battery Condition Technical Report

REPORT ID: BMS-REP-20261021-2667
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REPORTING PERIOD: 24H
BATTERY BUS: 12.0V Nominal

SPECIFICATION STATUS: UNVERIFIED PROTOTYPE · Profile: **Nominal 12V Lithium-Ion (Unverified Chemistry)** (Unverified Lithium-Ion). Exact chemistry and model are unverified. Provisional safety margins are applied. Do not exceed 45°C ambient or provisional voltage bounds.

SECTION A: OVERALL OBSERVED CONDITION

STATUS: INSUFFICIENT DATA

NOMINAL OPERATING STATE: Measurements are within verified manufacturer limits (Nominal 12V Lithium-Ion (Unverified Chemistry)). Thermal dissipation is balanced with no detected anomalous temperature rise.

SECTION B: LATEST VALID SENSOR MEASUREMENTS

Directly sampled from hardware registers via Firebase Realtime Database (/smart_bms/latest).

Parameter	Measured Value	Hardware / Path	Data Status	Technical Context / Scope
Bus Voltage	**11.15 V**	INA219 (I²C 0x40)	**VALID**	Battery terminal potential across high-side shunt.
Shunt Current	**0.108 A**	INA219 (I²C 0x40)	**VALID**	Positive: Charge / Negative: Discharge across 0.1Ω.
Active Power	**1.16 W**	Calculated (V × I)	VALID	Real-time electrical power transferred to load bus.
Ambient Temperature	**29.8 °C**	DHT11 (1-Wire D7)	**VALID**	**Enclosure ambient air** near pack (NOT cell core).
Ambient Humidity	**47.1 %**	DHT11 (1-Wire D7)	VALID	Relative air humidity inside prototype enclosure.

SECTION C: KEY FINDINGS & OBSERVATIONS

1. Monitored nominal 12V bus voltage recorded at 11.15 V.
2. Ambient enclosure temperature measured at 29.8 °C; cooling fan threshold at 35.0°C.
3. No immediate hardware safety trip conditions detected in latest sample.

SECTION D: HISTORICAL TRENDS & STATISTICAL EXTREMA

Derived mathematically over 70 sampled interval(s) for the reporting period.

Monitored Channel	Minimum	Maximum	Average	Valid Samples	Observation / Stability
Bus Voltage	0.01 V	11.22 V	**7.83 V**	70	Bus potential within monitored bounds.
Load Current	0.00 A	0.11 A	**0.08 A**	70	Current draw across INA219 shunt.
Ambient Temperature	29.00 °C	32.20 °C	**31.02 °C**	70	DHT11 thermal buildup in enclosure.

SECTION E: EVIDENCE-BASED RECOMMENDED ACTIONS

#	Action Required	Technical Justification & Verification Target
1	**Continue continuous CAN telemetry and thermal monitoring.**	Adhere to battery manufacturer technical specifications and safety envelopes.
2	**Note: Normal resting voltage alone does not prove long-term cell health; maintain periodic capacity testing.**	Adhere to battery manufacturer technical specifications and safety envelopes.

SECTION F: DATA QUALITY & SENSING LIMITATIONS

1. **Ambient vs. Core Thermal Measurement:** The DHT11 sensor measures enclosure air temperature adjacent to the pack. It does NOT capture internal electrochemical core cell temperature or localized thermal hotspots.
2. **Unverified Chemistry Constraints:** Exact cell chemistry and manufacturer model have not been laboratory-verified. Consequently, manufacturer-specific C-rate curves, SOC, SOH, and RUL are intentionally withheld to prevent misleading claims.
3. **Single-Node Acquisition:** Telemetry is sampled across I²C (INA219) and 1-Wire (DHT11) on the NodeMCU ESP8266 controller. Stale readings (> 60 seconds) are automatically flagged as unverified.

SECTION G: RETRIEVED TECHNICAL CITATIONS

• Texas Instruments — INA219 Bidirectional Current and Power Monitor Datasheet (SBOS448G)
• Aosong Electronics — DHT11 Humidity & Temperature Digital Sensor Technical Specification
• IEEE Std 1491-2012 — Guide for Selection and Use of Battery Monitoring Equipment in Stationary Applications
• ISO 11898-2 — Road Vehicles — Controller Area Network (CAN) — Part 2: High-Speed Medium Access Unit